P. Centrifugal casting
Q. Pressure die casting
R. Investment casting
S. Shell mould casting

2. Carburetors
3. Gear blanks
4. Hollow pipes

Codes:

	P	Q	R	S
(a)	4	2	1	3
(b)	4	1	2	3
(c)	3	4	2	1
(d)	1	2	3	4

Solution: (a)

EXAMPLE : 1.15 The pattern material used for producing wave guides of radar system using casting process will be

(a) Plaster of paris
(b) Frozen mercury
(c) Wax
(d) Polystyrene

Solution: (c)

Due to complex shape of the casting and possibility of reusability of pattern wax pattern with investment casting will be used for producing wave guides for radar system.

1.25 CASTING DEFECTS

1. ***Drop***

Irregular projection on the top of casting caused by dropping of sand from cope.

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2. Buckle

V-shaded depression occurring on flat casting due to expansions of sand at the mould face before liquid metal solidifies.



Figure: Buckle

3. Scab

Protruding surface of casting at roof.

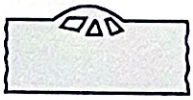


Figure: Scab

4. Swell

Liquid metal displaces the sand at the wall regions due to hydrostatic pressure.



Figure: Swell

5. Mould Shift

Due to misalignment between the two halves.

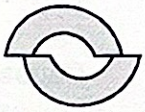


Figure: Mould Shift

6. Discontinuities

Discontinuities is defined in casting in terms of cracks, hot tearing and cold shut. If the solidifying metal is constrained from shrinking freely, cracking and tearing can occur. Although many factors are involved as tearing, coarse grain size and the presence of low melting point segregates along the grain boundaries, increase the tendency for hot tearing. Cold shut is an interface in a casting that lacks complete fusion because of the meeting of two streams of liquids metal from different gates.

7. Misrun

Due to lack of fluidity and pouring temperature before filling of the metal in the cavity it solidifies.

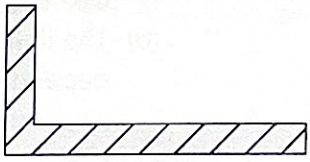


Figure: Misrun

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8. Inclusions

It form during melting, solidification and molding. Generally nonmetallic, they are regarded as harmful because they act as stress raisers and reduce the strength of casting. Inclusion may form during melting when molten metal reacts with the environment or with the crucible or mold material. Chemical reaction among components in the molten metal may produce inclusion slag and other foreign material entrapped in the molten metal also become inclusion.

EXAMPLE : 1.16

The casting defect in which rough spots of excess metal appears because of erosion of moulding sand is known as

(a) Drop
(b) Swell
(c) Metal penetration
(d) Cuts

Solution: (d)

Cuts and washes : If velocity of molten metal is high or strength of mould is less then it results in erosion of moulding sand and results in appearance of rough spots.

EXAMPLE : 1.17

Which of the following pairs of casting process & applications are correctly

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